International Journal of Fatigue 144 (2021) 106044

Contents lists available at ScienceDirect

International Journal of Fatigue

journal homepage: www.elsevier.com/locate/ijfatigue

Stochastic approach to fatigue crack-growth simulation for a railway axle under input data variability

V. Giannella

Department of Industrial Engineering, University of Salerno, via Giovanni Paolo II, Fisciano (SA), Italy

ARTICLE INFO

Keywords:
Fatigue crack-growth
Railway axle
Life prediction
Uncertainty
NDT inspections

ABSTRACT

Fatigue crack-growth of mechanical components is stochastic in nature and deterministic approaches do not allow for its complete study. Primary sources of this stochasticity can be: the inherent variability of material properties, the nature of service load fluctuations, the approximations assumed by the algorithms to calculate the main fracture parameters.

This work reports a stochastic approach to the fatigue crack-growth in a railway axle under variability of different input parameters. The approach is used to predict the probability distribution of the residual fatigue life of a railway axle in presence of defects taking into account multiple sources of uncertainty.

1. Introduction

The causes of catastrophic engineering failures can be often attributed to a combination of material deficiencies in the form of pre-existing flaws in the material, poor designs, in-service damages, etc. To ensure safety, current specific standards require routine periodic checks for detecting possible cracks. Cracked components have to be monitored and, if necessary, replaced or repaired before they become critical. Improvement at the design stage, where high stress concentrations in the structure should be avoided, better production methods as well as enhancement of material properties, have all helped to minimize the criticalities and consequently reducing the amount of these failures. However, the elimination of cracks is not only impractical but also impossible because cracks often nucleate well below the material yield strength. To further mitigate fracture failures, the "damage tolerance" philosophy has been introduced in recent years upfront during the design stage, when engineers have to anticipate the likelihood of crack nucleation in the structural components. As structures are becoming more complex, the need for an accurate and reliable assessment of the structural safety has become even more crucial, and a simple arbitrary safety factor is no longer an acceptable safety margin, nor it is justified in terms of economy and/or efficiency. Linear Elastic Fracture Mechanics (LEFM) is often used to provide the necessary additional safety checks, understandings of the fracture processes and, even more importantly, obtaining reliable predictions on residual strength of structures [1].

Since early studies, it was understood that the crack-growth phenomenon in mechanical components is stochastic in nature and safety factors were introduced to counterbalance the effects of multiple sources of uncertainty. Unfortunately, the direct use of these safety factors often resulted in conservative or even over-conservative results; this highlighted that deterministic approaches to fracture were limiting methods to tackle these kind of phenomena. To this aim, empirical and semi-probabilistic approaches have been suggested in the standards/codes, providing the designer with a series of partial safety factors accompanying with limit state design codes to achieve the target reliability [2]. Among them, Eurocode 3 [3] introduced S-N diagrams under 5% failure probability and suggested a safety factor to be used onto the diagram. Also the nuclear design code [4] defined that fatigue curves of material have to be shifted by either a factor of two on stresses/strains or a factor of twenty on the predicted lives. These factors were not defined as safety margins but rather as adjustments for the uncertainties and scatter resulting from the laboratory fatigue test data to those of the actual components [5].

Until now, continuous efforts have been made for the purpose of improving the accuracy of the fatigue crack-growth predictions. Among them, deterministic approaches have been widely demonstrated to be capable of providing accurate predictions even in complex operating conditions [6], with particular load spectra [7], etc. These advanced deterministic approaches have been widely used in the industry, even if they lead to predictions that do not provide any adequate information about their reliability for the actual structures. Moreover, design codes and standards are generally empirical, too conservative and also difficult to adapt even to slight changes on materials and processes. Therefore, the need for designing processes that provide life predictions of mechanical components with well-defined ranges of variability, nowadays

E-mail address: vgiannella@unisa.it.

https://doi.org/10.1016/j.ijfatigue.2020.106044
Received 9 August 2020; Received in revised form 4 November 2020; Accepted 6 November 2020
Available online 24 November 2020
0142-1123/© 2020 Elsevier Ltd. All rights reserved.

Nomenclature

A_i	*i-th* constant used for calculation of f
a	crack depth
a_0	El-Haddad's parameter
C	coefficient of NASGRO formulation
C_{th}^{+}	parameters for calculation of ΔK_{th} for positive R values
C_{th}^{-}	parameters for calculation of ΔK_{th} for negative R values
D	diameter of axle
D_{rw}	diameter of railway wheel
da/dN	crack-growth rate
f	Newman's crack closure coefficient
r	distance from crack tip
m	exponent of NASGRO formulation
N	number of cycles
n	prediction counter
K	stress intensity factor
K_c	critical stress intensity factor
$K_{bending}$	stress intensity factor due to bending only
K_{max}	maximum stress intensity factor
K_{min}	minimum stress intensity factor
K_{op}	opening stress intensity factor
$K_{press-fit}$	stress intensity factor due to press-fit only
N_{LS}	mileage per load spectrum
P_f	probability to cover all the possible combinations
p	exponent of NASGRO formulation for near-threshold regime
q	exponent of NASGRO formulation for sub-critical regime
R	stress ratio
$RMS_{\frac{da}{dN}}$	Root Mean Square error in terms of da/dN
$RMS_{\Delta K_{th}}$	Root Mean Square error in terms ofΔK_{th}
s	number of iterations
s_{max}	maximum applied nominal stress in NASGRO formulation
α	constraint factor
Δa	crack advance
ΔK	range of stress intensity factor
ΔK_{th}	R-dependent threshold stress intensity factor range
$\Delta K_{th,0}$	threshold stress intensity factor range for R = 0
$\Delta N_{i,i+1}$	number of cycles between two consecutive propagation steps
σ_0	flow stress
σ_{max}	applied maximum nominal stress
σ_{ys}	yield stress
σ_{UTS}	ultimate tensile strength
DBEM	Dual Boundary Element Method
CGR	Crack-Growth Rate
FEM	Finite Element Method
LEFM	Linear Elastic Fracture Mechanics
MTS	Maximum Tangential Stress
MSED	Minimum Strain Energy Density
RMS	Root Mean Square

appears to be one of the key aspects of mechanical design.

Until now, several studies have explored the possibility to choose safety factors from a probabilistic design perspective. Most of the available research focused primarily on obtaining the life prediction by considering few variations of the given loading conditions as well as variations on the crack evolution and growth rates [8–10]. According to this, the purpose of this study was to quantify the effects of the multiple sources of uncertainty on the phenomenon of fatigue crack-growth [11]. The considered case study was a railway axle whose accurate designing process is still nowadays of great importance for the railway industry, as their failures may cause serious accidents [12–14]. The design of these components is regulated by technical recommendations, which essentially define the maximum permissible fatigue stresses and the corresponding diameters for the axles [15,16]. Beside the classical approaches, there is an increasing demand for the reliability assessment of the railway axles, especially considering the recent applications for high-speed trains. Advanced deterministic approaches were recently proposed to tackle the problem of fatigue crack-growth in the axles, providing results in terms of residual mileage due to the presence of cracks and reasonable inspection intervals to use for maintenance [17,18]. Nevertheless, in the safety design of these critical components of mechanical systems, the various sources of uncertainty including variations in material properties, loading and geometry, have to be taken into account. Some examples of probabilistic approaches to tackle the problem of fatigue crack-growth in railway axles can be found in [19,20]. In particular, [20] proposed a stochastic approach to fatigue crack-growth similarly to what presented here, even if it considered that only two material parameters (C and $\Delta K_{th,0}$) were effective sources of variability, thus without any consideration on the actual loading conditions, the fracture algorithms accuracy, environmental effects, etc. On the other hand, the framework presented in the current investigation can be considered as similar to that proposed in [21], even though the latter was related to a multiaxial fatigue case study. The main objective of this work was not the thorough fracture analysis related to a particular case study, but rather the comprehension of how residual fatigue life predictions are affected by the related input parameters. These parameters were seen as input sources of uncertainty and the quantification of their impact on the residual fatigue life of a railway axle was the main goal of this analysis.

The primary sources of uncertainty here considered were: i) the inherent scattering of material properties, ii) the statistical nature of service load fluctuations experienced during the actual working conditions of trains, iii) the accuracy of the algorithms adopted to calculate the fracture parameters to be used for the life predictions. Environmental effects were not taken into account within this investigation. An aggressive environment, e.g. corrosive, is known to represent a trigger for a diffuse crack nucleation at micro-scale (<0.3 mm), that in turn produces large cracks (greater than2 mm) by a mechanism of coalescence [22]. Since this investigation considered the propagation of a relatively large crack, i.e. having a 2 mm depth, the contributes coming from corrosion were considered as neglectable.

The rest of the paper is organised as follows. Section 2 develops how uncertainty sources was introduced in a fatigue crack-growth law to take into account of material properties variability. Section 3 develops how the variability given by different methods used to calculate the fracture parameters was taken into account. Section 4 develops how load fluctuations given by the working conditions of the train were considered as a source of uncertainty. Section 5 reports the overall description of the procedure comprising the stochastic input parameters previously described. Section 6 reports the results of the analyses together with some comments and recommendations. Finally, Section 7 concludes the remarks of the article.

2. Uncertainty in material properties

Fatigue cracking process, both nucleation and propagation, is inherently stochastic due to material variability, micro-structural irregularities, and uncontrolled actual loading conditions [19–23]. In addition, various sources of uncertainty arising from a simplified representation of actual physical process and/or sparse information on manufacturing, material properties and loading profiles contribute to the stochastic behaviour of the failure mechanisms [19–20]. Clear

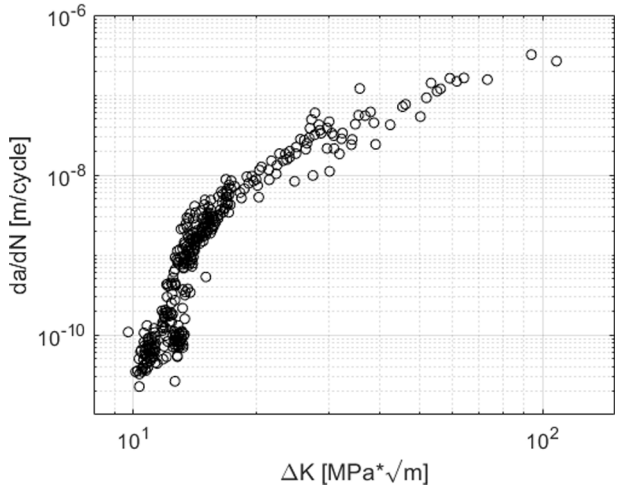


Fig. 1. 30NiCrMoV12 CGRs obtained from full-scale axles and specimens test data.

evidence of these aspects can be observed in test data related to low cycle fatigue, high cycle fatigue and also for fatigue Crack-Growth Rates (CGRs).

Considering that the present investigation focused on a railway axle, the material under analysis was the high-strength steel 30NiCrMoV12, according to the standard UNI 6787 [24] and representing one of the most accepted materials for these mechanical components. The related CGRs were available from literature [17] for both specimens and full-scale axles and are here reported in Fig. 1. The chemical composition and the main mechanical properties of 30NiCrMoV12 high-strength steel are reported in Table 1. It is worth noting that the data exhibit a significant scatter, in other words, a given ΔK leads to a series of possible CGRs, eventually resulting in a conspicuous scatter of the related life estimates. It is also worth recalling that CGRs should be calculated with the highest possible accuracy since they represent the fundamental parameter for the life assessment.

These test data are generally adopted in order to calibrate a fatigue crack-growth law useful to perform life estimates for cracked components made of the same material. Using the experimental data reported in Fig. 1, two crack-growth laws were calibrated in [17]. Within this document, that NASGRO curve will be termed as a "nominal" fitting of the NASGRO equation, whereas the same NASGRO formulation will be here taken into account in a probabilistic way. Such a NASGRO formulation was selected since it allowed to take into account of: i) the three distinct parts of the CGR diagram (near-threshold, Paris' range and critical zone); ii) the crack closure effects, according to Newman definition [25]; iii) the stress ratio influence on CGRs. More details on the NASGRO formulation are provided in the followings.

2.1. NASGRO formulation description

Different elements of the NASGRO formulation were initially documented in [26] and continuously developed along the years [26–27]. The main equation is reported in Eq. (1):

$$\frac{da}{dN} = C\left[\left(\frac{1-f}{1-R}\right)\Delta K\right]^m \frac{(1-\Delta K_{th}/\Delta K)^p}{(1-K_{max}/K_c)^q}, \tag{1}$$

where N is the number of applied fatigue cycles, a is the crack depth, da/dN in turn represents the CGR. R is the stress ratio, that for alternating bending is equal to $\sigma_{min}/\sigma_{max} = K_{min}/K_{max} = -1$. ΔK is the stress intensity factor range, equals to $K_{max} - K_{min}$.

C and m are empirical constants, respectively, used to modify shift and slope of the CGR curve in the intermediate part (Paris' range) of the CGR diagram. p and q are two empirically constants used to drive the shape for near-threshold (at low ΔK) and subcritical (at high ΔK) regions of the CGR curve respectively.

The so called crack opening function f allows to consider that the ΔK contributes only partially to the advancing of the crack. The phenomenon of crack closure was first detailed in [28], observing that the stiffnesses of cracked and non-cracked specimens were comparable for relatively low tensile loads. This led to the definition of a K opening $K_{op} > K_{min}$ representative of the minimum K value for which opening of crack faces really occurs. Crack closure phenomenon helped explaining a wide range of fatigue data, and has been especially important in the understanding of the effect of stress ratio R (less closure at higher R) and short cracks (less closure than for long cracks) [28,29]. For plasticity-induced crack closure, the parameter f was originally proposed in [25] as:

$$f = \frac{K_{op}}{K_{max}} = \begin{cases} max\left(R, A_0 + A_1 R + A_2 R^2 + A_3 R^3\right), & R \geq 0 \\ A_0 + A_1 R, & -2 \leq R < 0 \end{cases} \tag{2}$$

where the coefficients A_i were given by:

$$\left.\begin{aligned} A_0 &= \left(0.825 - 0.34\alpha + 0.05\alpha^2\right)\left[\cos\left(\frac{\pi}{2} s_{max}/\sigma_0\right)\right]^{1/\alpha} \\ A_1 &= (0.415 - 0.071\alpha) s_{max}/\sigma_0 \\ A_2 &= 1 - A_0 - A_1 - A_3 \\ A_3 &= 2A_0 + A_1 - 1 \end{aligned}\right\} \tag{3}$$

In Eq. (3), α is a plane stress/strain constraint factor, that is considered as physically meaningful when having values ranging from 1, which corresponds to a plane stress condition, to 3, which corresponds to a plane strain condition. It is generally assumed that relatively high α values (2.5–3) are assigned to materials such as the high-strength steel here considered, while low-strength materials usually have α values lower than 2.0.

The ratio s_{max}/σ_0 is the ratio of the maximum applied stress to the flow stress, the latter generally considered as the average value between the yield stress σ_{ys} and the ultimate tensile stress σ_{UTS}. Even though such ratio varies through the propagation and/or when considering load spectrum effects, it is generally assumed to be as a constant.

K_c is the critical K value for which final rupture of component occurs.

In order to take into account that the threshold stress intensity factor ΔK_{th} is significantly dependant from R, the following equation can be retrieved from literature:

$$\Delta K_{th} = \Delta K_{th,0} \frac{\sqrt{a/(a+a_0)}}{\left[\frac{1-f}{(1-A_0)(1-R)}\right]^{1+C_{th}R}}, \tag{4}$$

where $\Delta K_{th,0}$ is the ΔK_{th} corresponding to $R = 0$. C_{th} is a simple fitting parameter for test data, having a positive value for positive R and a negative value for negative R.

Finally, the El-Haddad's parameter a_0, originally proposed in [30], allows to take into account of the crack-growth for short cracks, namely

Table 1
Chemical composition of 30NiCrMoV12 (minimum-maximum concentrations).

C	Si	Mn	Pa	S	Cr	Cu	Mo	Ni	V
0.26		0.40			0.60		0.40	2.70	0.08
0.32	0.40	0.70	0.020	0.015	1.00	0.20	0.60	3.30	0.13

Table 2
NASGRO nominal parameters for 30NiCrMoV12.

σ_{UTS} [MPa]	σ_{ys} [MPa]	K_c [MPa $\sqrt{m}$]	C [MPa^{-n} m$^{1-n/2}$]	m [–]	p [–]
980	880	120	1.15e−10	2.41	0.65
q [–]	a_0 [m]	C_{th}^{+} [–]	C_{th}^{-} [–]	$\Delta K_{th,0}$ [MPa $\sqrt{m}$]	α [–]
0.001	3.81e−5	1.4	−0.07	6.13	2.5

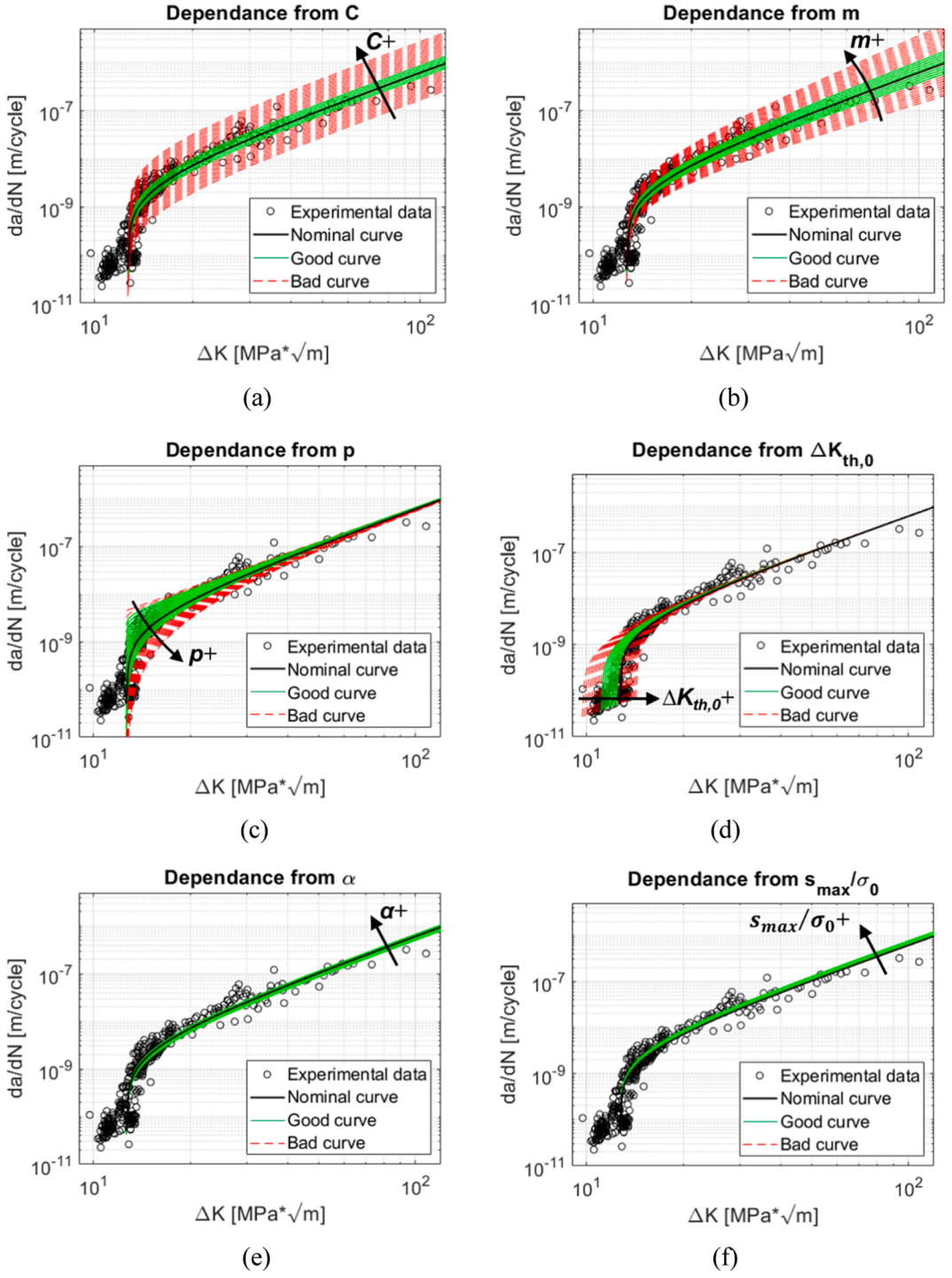


Fig. 2. CGR curve variability produced by the main NASGRO parameters; experimental data and nominal curve are also shown; "Good" and "Bad" curve refers to RMS errors lower/higher than cut-off values respectively.

Table 3
Ranges of variability for the main NASGRO material data.

	Min value	Nominal value (Table 2 data)	Max value
$Log_{10}(C)$ (C [MPa^{-m} m$^{1-m/2}$])	−10.5	−9.9393	−6.6
m [–]	0.1	2.410	2. 9
p [–]	0.2	0.650	4
$\Delta K_{th,0}$ [MPa $\sqrt{m}$]	4.85	6.13	6.05

when crack sizes are comparable with the grain size of the material. Even though a_0 varies with the material, it is generally always assumed to be equals to 0.0381 mm, see [17,25].

Data required by the NASGRO formulation were available from the literature [17] or ad hoc calculated. These data are here referenced as "nominal" parameters and are listed in Table 2, whereas the stochastic approach to the formulation will be discussed below.

2.2. Stochastic approach to NASGRO formulation

By using data and equations as those above described, fatigue life predictions in presence of defects can be computed. This is generally performed in a deterministic way, even though such predictions would not provide any satisfactory range of variability, thus implying the introduction of safety margins that inevitably lead to conservative designing choices. To better capture the uncertainty given by the variability of material properties, the main parameters C, m, p, q, $\Delta K_{th,0}$, α, s_{max}/σ_0 required by the NASGRO formulation were here considered as uncorrelated random variables assuming values included in given ranges.

To this aim, a series of CGR curves were drawn in order to evaluate the influence of each NASGRO parameter on the final curve shape. Fig. 2 shows the CGR curves obtained by varying one of the main parameters at a time, while keeping the remaining ones as fixed to their nominal values (Table 2). In order to evaluate the quality of the n-th CGR curve, two Root Mean Square (RMS) errors were computed with Eq. (5) for both the near-threshold regime and the Paris' range. Both conditions in Eq. (5) had to be satisfied by the generic n-th CGR curve. Namely, CGR curves were defined as either "Good" or "Bad" if the related RMS errors in terms of both da/dN(for Paris' range) and $\Delta K_{th,0}$ (for near-threshold regime) were lower than user defined cut-off values. Due to the different amount of scattering of experimental data in the two areas of interest, different cut-off values were defined. A maximum RMS error of 3% was allowed for Paris' range, whereas 4% was considered for the near-threshold regime. This procedure allowed to quantify the deviation between each single CGR curve and the full set of experimental data, allowing to estimate the accuracy of each simulated CGR curve in fitting the experimental data.

$$\begin{cases} RMS_{\frac{da}{dN},n} = \sqrt{\dfrac{\sum_n^d \left\{ \left[\dfrac{\log\left[\left(\frac{da}{dN}\right)_{exp}\right]}{\log\left[\left(\frac{da}{dN}\right)_n\right]} - 1 \right]^2 \right\}}{d}} < 3\% \\ RMS_{\Delta Kth,0,n} = \sqrt{\dfrac{\sum_n^d \left\{ \left[\dfrac{\left(\Delta K_{th,0}\right)_{exp}}{\left(\Delta K_{th,0}\right)_n} - 1 \right]^2 \right\}}{d}} < 4\% \end{cases} \quad (5)$$

In Eq. (5), d is the total amount of experimental data, whereas

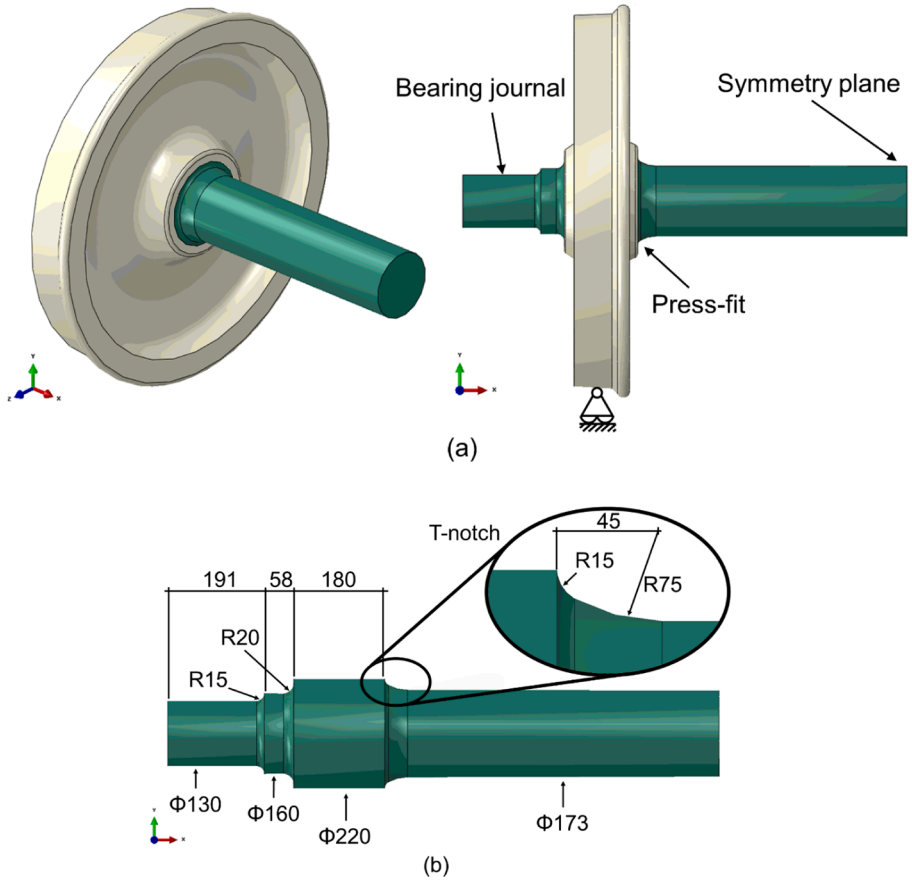


Fig. 3. (a) CAD model of the wheel-axle assembly (b) and highlight of the main dimensions.

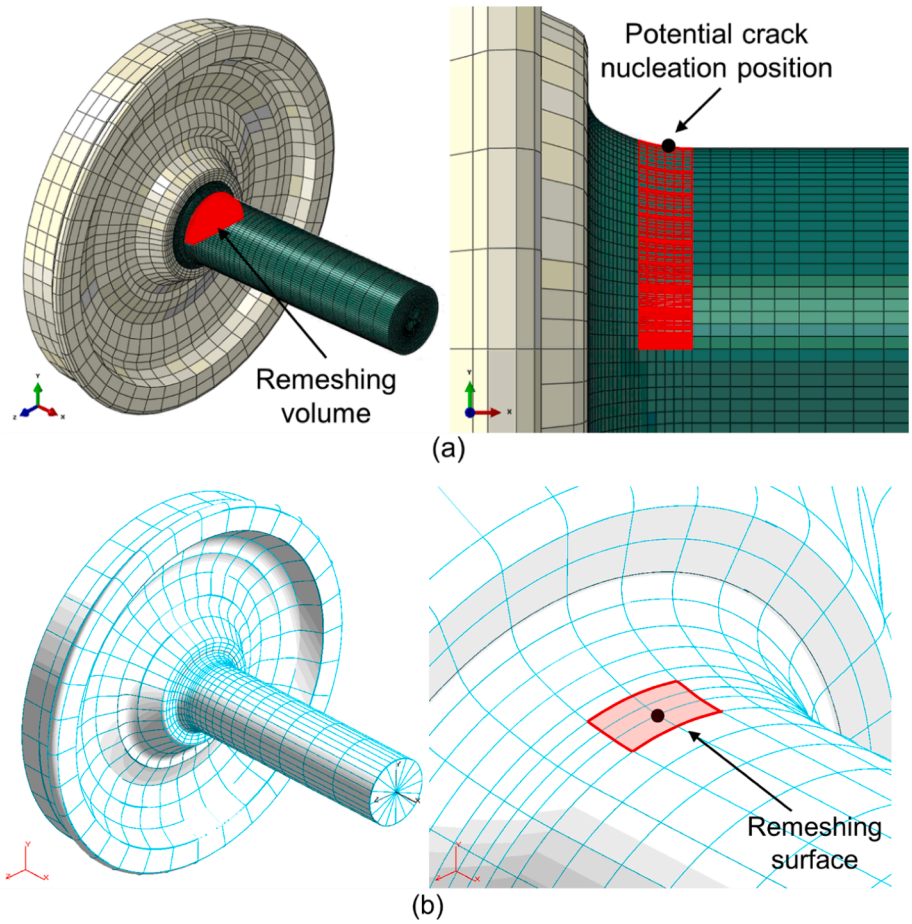


Fig. 4. (a) FEM and (b) DBEM models with highlights of the crack insertion positions (black dots) and remeshing FEM volumes and DBEM surfaces (in red). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$RMS_{\frac{da}{dN},n}$ and $RMS_{\Delta Kth,0,n}$ represent the errors of a n-th CGR curve in terms of da/dN and $\Delta K_{th,0}$ respectively.

The parameter q was not considered as variable in this investigation since the material did not present a significant increment of the CGR in the critical zone (at high ΔK; Fig. 1). Nevertheless, negligible residual fatigue crack propagation is calculated in the critical zone, thus proving the general validity of the current investigation. With reference to Fig. 2e-f, it can be noticed that parameters α and s_{max}/σ_0 do not modify the CGR curve significantly, therefore, their values were considered as fixed to the nominal ones for the subsequent analyses. This allowed to reduce the number of parameters to consider from the stochastic standpoint, with a consequent reduction of the overall computational runtimes.

Finally, the parameters C, m, p, $\Delta K_{th,0}$ were considered as the main parameters that influence the CGR curve shape. These parameters were considered as continuously varying in given ranges (Table 3), these latter selected as large as possible so as to enclose all the possible occurrences of the parameters. A more detailed discussion on this aspect is provided in Section 6.

3. Uncertainty in fracture algorithms

Numerous crack-growth algorithms exist in literature to describe the fracture process that occurs in engineering structures. Even if empirical/semi-empirical formulations and analytical approaches exist, they are generally limited to simple geometries and/or operating conditions [31,32]. Hence, the numerical simulation has become the only existent tool that allows to cope with the continuously increasing demands coming from the industry. Several numerical methods have been demonstrated as being effective to simulate fracture phenomena; nowadays, the most widespread are the Finite Element Method (FEM), thanks to its versatility [33–35], and the Dual Boundary Element Method (DBEM), thanks to its inherent efficiency in dealing with fracture problems [36–49]. It is worth mentioning that numerous alternative numerical methods have been recently proposed but, even if promising, they seem to be still in an early stage of their development, e.g. meshless methods [40] and peridynamics [41].

The current investigation focused on a railway axle under rotary bending condition, for which two numerical methods were used. In particular, FEM and DBEM numerical methods were adopted and the related results cross compared. These calculations were performed in order to acquire a sensibility on the results accuracy when selecting one method over another. All the details are provided in the followings.

3.1. Numerical analyses

Two numerical simulations were separately developed with the FRANC3D [42] code, adopted as a FEM fracture tool, and with BEASY [43] code as DBEM code specialised for fracture.

Both numerical calculations were performed by considering the geometry, shown in Fig. 3, of a non-powered axle of a Y25 bogie. This geometry was taken from literature [44] and considered here as a case study. It consisted of a bore axle coupled with a railway wheel, having average outer diameter D_{rw} equals to 940 mm. Only half axle was modelled assuming a longitudinal symmetry. Loading conditions consisted of a bearing load, inducing a maximum nominal flexural stress equal to σ_{max}, coupled to a press-fit condition, with radial interference fit equals to 125 μm (according to the standards [44,45]). The bending

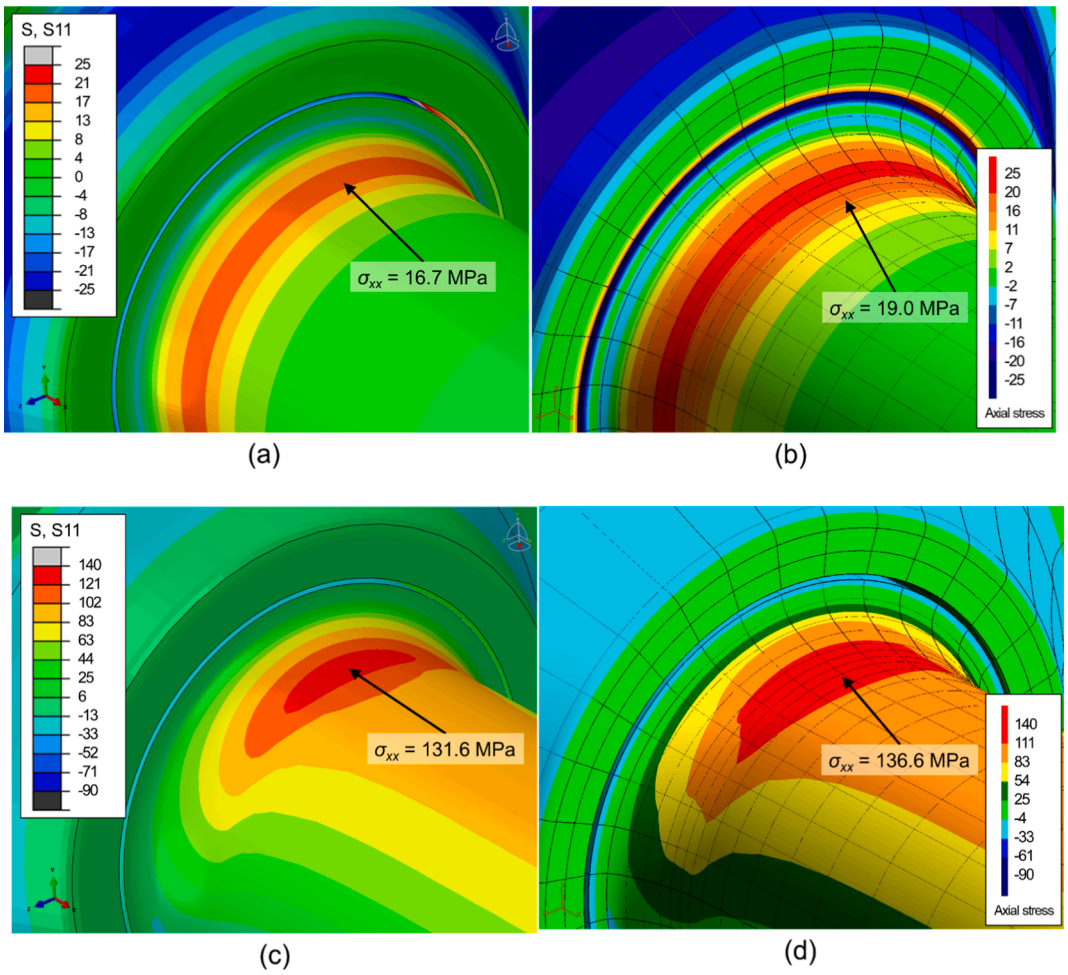


Fig. 5. (a, c) FEM and (b, d) DBEM axial stresses at T-notch due to press-fit only (a-b) and due to press-fit plus bearing load (c-d).

condition was considered as alternating rather than rotary, as also made in similar works [17–20], whereas the press-fit was considered as constantly applied to the axle, thus leading to a stress ratio $R \neq -1$. It is worth noting that the press-fit induces a tensile axial stress on the outer surface at the T-notch (Fig. 3), whereas compressive stresses are generated internally. Considering also that cracks nucleate at the most critical locations on the outer surface, these cracks at T-notch undergo a propagation with variable R. This aspect was taken into account by calculating R with Eq. (6):

$$R = \frac{K_{min}}{K_{max}} = \frac{K_{press-fit} - K_{bending}}{K_{press-fit} + K_{bending}} \tag{6}$$

R ranged between −0.70 and −1.04 depending on the crack size and on the actual loading conditions considered by the load spectrum. Moreover, the superposition of press-fit stresses to the bending ones resulted in a reduction of the fatigue life estimates by a factor of nearly three, since press-fit increased K_{max} due to bending only and reduced also ΔK_{th}.

The material of the axle was the widely used high-strength steel 30NiCrMoV12, according to UNI 6787 [24], with Young's' modulus equals to 200 GPa and Poisson' ratio equal to 0.3. These material data were not considered as random variables since they do not influence the fracture parameters calculated under LEFM assumptions.

Regarding the FEM model (Fig. 4a), it comprised nearly 120k 8-noded and 20-noded hexahedral elements, this number increased to 220k along with the advancing of the crack. In order to insert and propagate the crack, a step-by-step remeshing procedure was chosen, implying a mesh refinement only in a small volume enclosing the crack (in red in Fig. 4a), while keeping as fixed the bulk of the remaining mesh. Crack front was enclosed by a pipe of "wedge" 15-noded elements and "quarter-point" elements at crack tip (to better capture the $1/\sqrt{r}$ crack tip stress and strain singularity [46]). Finally, the remaining remeshed volume was filled with tetrahedral 10-noded elements. Further details about such a simulation procedure can be also found in [34,35,42,50].

Regarding the DBEM model (Fig. 4b), it comprised nearly 2100 4-noded quadrilateral surface elements for the uncracked model, reaching nearly 5100 elements for the last propagation steps. As performed for FEM simulations, a step-by-step localised remeshing procedure was also adopted by DBEM, considering a localised remeshing surface (in red in Fig. 4b) constantly increasing in size with the enlarging of the crack. 8-noded quadrilateral and 6-noded triangular surface elements were used to fill the remeshing surface and a "structured" line of elements was always guaranteed along the crack tip. Discontinuous elements were used to mesh both crack faces. Rings of internal points were properly positioned along crack fronts in order to calculate the J-integral, in turn to get the three separate K values. Further details about this procedure can be found in [35,36,43,48,49].

Contour plots of the axial stresses at the T-notch were reported in Fig. 5 for both uncracked models. A total vertical load of 20 tons was applied on the bearing journal for such a purpose.

Crack-growth simulations were setup as incremental approaches that, starting from a crack having a given user defined size and shape, allowed to simulate the crack-growth by consecutive discrete steps of propagation until the critical crack size was reached. A transversal semi-circular crack with radius equal to 2 mm was arbitrarily assumed as nucleated at the T-notch, see Figs. 4 and 5. Such initial crack was

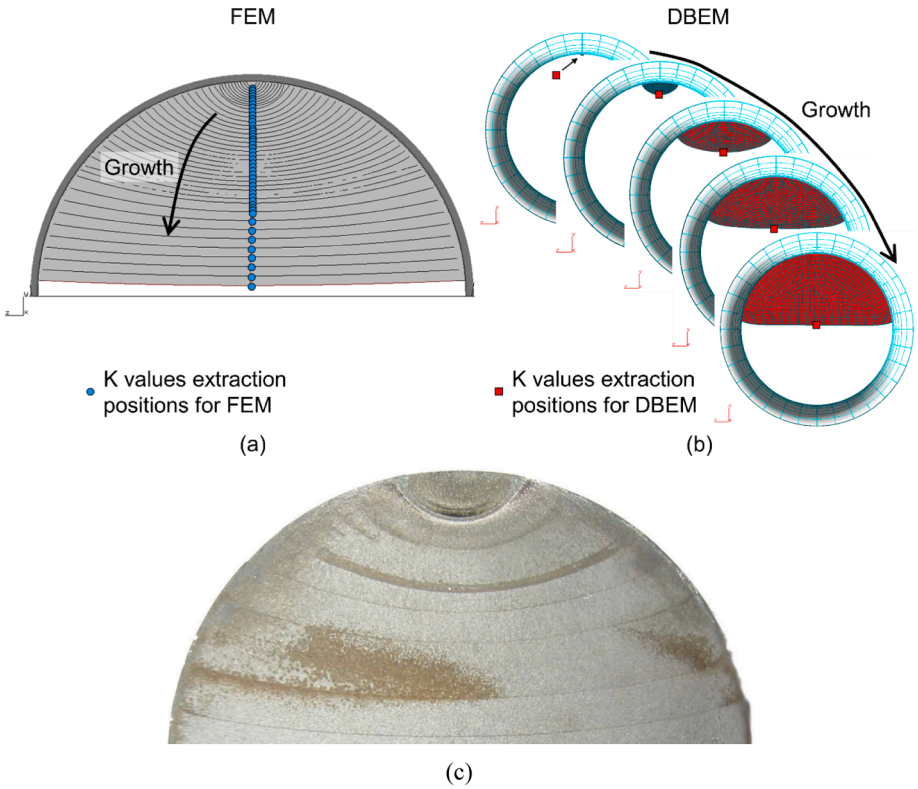


Fig. 6. Crack shapes computed by (a) FEM and (b) DBEM; experimental evidence of fatigue crack-growth in a railway axle [54].

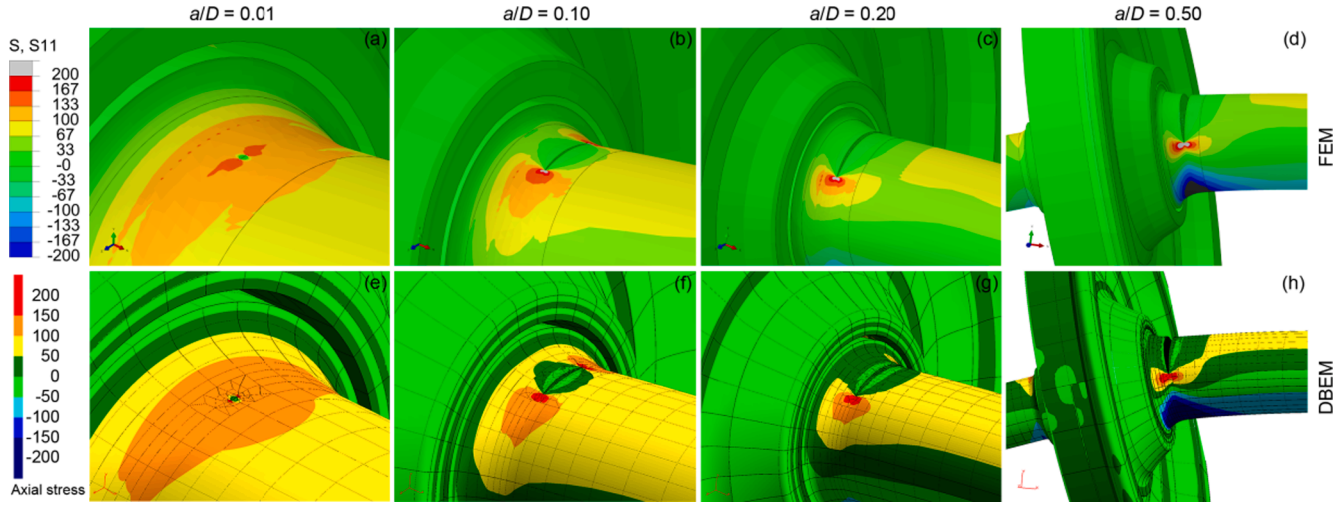


Fig. 7. Evolution of crack shapes for (a-d) FEM and (e-h) DBEM simulations including axial stress contour plots (min/max levels −200 MPa/+200 MPa, deform scale = 50).

advanced step-wise through the cross-section of the axle in several incremental steps with average crack advances per step $\Delta a_{i,i+1} = a_{i+1} - a_i$ ranging between 1 and 4 mm. Even if performed with different algorithms, both FEM and DBEM crack-growth procedures consisted in an iteration of the following sub-steps:

a) crack insertion: the *i*-th crack is inserted in the model, involving a local rearrangement of the surrounding mesh in order to accommodate the *i*-th crack geometry. Localised rearrangements of the meshes were used to retain coarser meshes far from the crack, consequently reducing runtimes (Fig. 4).

b) model solving: the *i*-th cracked model is solved, the stress-strain maps as well as *K* factors along *i*-th crack front are calculated. *K* were calculated by using the *J*-integral technique [43,48,49] and the *M*-integral technique [42,50] for DBEM and FEM respectively.

c) crack extension: the *(i + 1)*-th crack front is predicted by considering the *K* factors calculated along *i*-th crack front and the model geometry. Kinking of the crack was predicted by the Minimum Strain Energy Density (MSED) [51,52] and the Maximum Tangential Stress (MTS) [53] criteria for DBEM and FEM respectively. Eq. (7) was used to calculate the nodal advancing per step along each crack front for both methods:

$$\Delta a_x = \Delta a_y \left(\Delta K_x / \Delta K_y\right)^m, \tag{7}$$

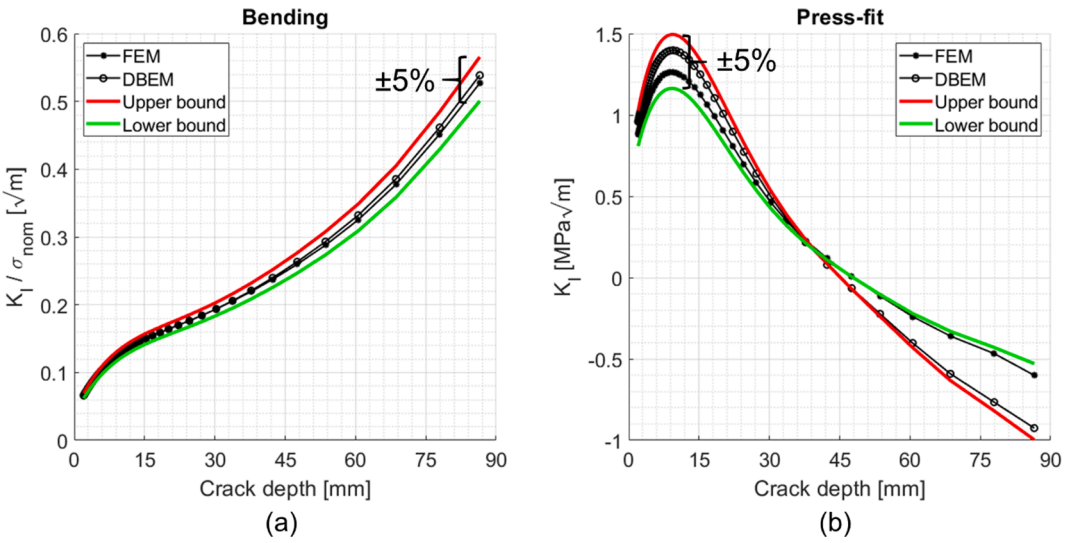


Fig. 8. Series of K_I for mid-side point along crack front (crack deepest point) for loading conditions of (a) bending and (b) press-fit.

where Δa_x and ΔK_x represent the advance per step and the K range for the generic x-th crack front node respectively. In order to obtain a realistic advancing of the crack, for each crack advancing from i-th to $(i + 1)$-th, it must be guaranteed that the same fatigue life prediction is estimated for all the advancing front nodes. Even if, strictly speaking, Eq. (7) is consistent with a Paris' law rather than a NASGRO, this equation is very often assumed by the codes since producing negligible approximations [42,43]. Further applications of these procedures can be found in [33–39,47].

Fig. 6a-b and Fig. 7 show the advancing of the cracks through the cross sections of the axles for both FEM and DBEM modelling methods. A very good agreement was obtained among the numerical crack shapes and a typical experimental evidence (available in [54]). K_I values were the only effective contributors to crack-growth since K_{II} and K_{III} were negligible for most of the propagation. This was expected since the main stress contributor was the bending, whereas the press-fit played a role only initially, i.e. when the related axial stresses (~15 MPa) were comparable with the bending ones (50–180 MPa). It is also worth to observe that the notch effect, that affects the outer surface primarily, produces an increment of CGR on the outer surface, in turn resulting in a more "straight" crack front along with the propagation.

In Fig. 8, the K_I values calculated for the mid-side nodes of crack fronts (deepest point) were reported against the crack depths for both FEM and DBEM. Substantially, the bending loading condition produced nearly overlapped K values, whereas major discrepancies occurred when considering the press-fit condition, for which too demanding mesh refinements were needed. These discrepancies are expected to provide a certain impact on the life predictions since most of the residual fatigue life is predicted when the crack is relatively small (<20/30 mm), thus when press-fit condition turns out to be more relevant.

Furthermore, the relatively small discrepancies between curves of K values affect the life predictions because, due to the exponent m (Eq. (1)), a very small approximation on the calculation of the ΔK can eventually lead to large discrepancies on CGRs. It is also worth noting that also the same methodology could give slightly different results in terms of K. Numerical methods such as FEM or DBEM can generate slightly different K values, e.g. by varying the meshes, order or type of elements, criteria for calculating K and/or propagation angles, etc. Therefore, the definition of a range of variability for K values should be considered to take into account of the impact of this source of uncertainty on the residual fatigue life. To this aim, an upper and a lower bound for K vs. a curves were defined in such a way to consider a potential ±5% of variability for these data. During the extractions

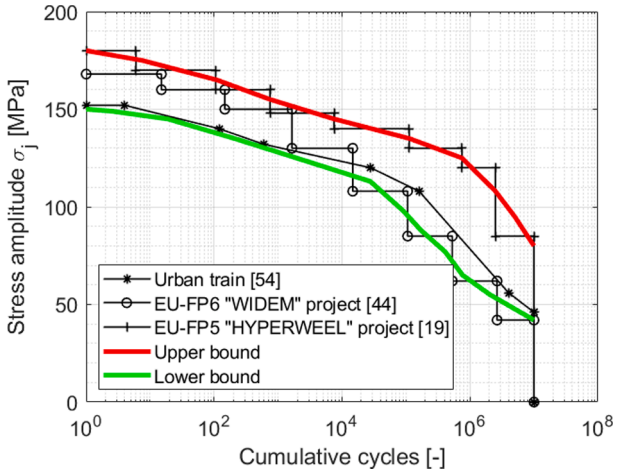


Fig. 9. On-board measured stress spectra and related upper and lower bounds.

requested by the stochastic procedure described below, two curves of Fig. 8, one for bending and one for press-fit, were randomly generated within their related ranges, thus trying to take into account of potential approximations in the calculation of K values. Details of this stochastic procedure are provided in Section 5.

4. Uncertainty on operating conditions

Nowadays, non-destructive inspection methods can detect, with relatively high probability of detection, only relatively long cracks (larger than 1–2 mm), depending upon the considered inspection technique [17,18]. Hence, there is the risk that existing cracks are not detected, so that conservative assumptions have to be made on the initial cracks to be modelled, in turn affecting the accuracy of the residual fatigue life estimations.

Railway axles are subjected to variable amplitude loading caused by the numerous different regimes of train operation. The dominant load is caused by the weight of the vehicle which generates rotary bending during train movement, while torsional stresses caused by the braking and motor systems have generally significantly lower magnitudes. Furthermore, there are additional forces which are generated when train goes through curved tracks, over crossovers, switches, rail joints etc. CGRs and crack paths depend on the load spectrum that the axle

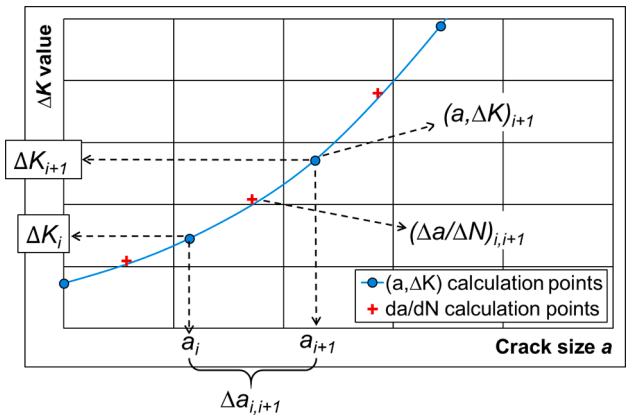


Fig. 10. Schematic view of main parameters for fatigue life prediction.

undergoes during its actual in-service operation. Consequently, for a reliable assessment of the residual fatigue life of a railway axle, it is necessary to know the real load spectrum that train undergoes. This is clearly impractical standing the stochastic nature of the data to measure.

In this work, the uncertainty on the axle loading conditions was tackled in the following manner. On the basis of load spectra obtained from on-board experimental measurements on high-speed and urban trains available in [19,54], two bounds were defined to enclose a range of variability of all the potential load spectra, see Fig. 9. Based on such a range of variability, a series of load spectra were drawn in between the upper and lower bounds. Consequently, for each extraction of the stochastic algorithm, a load spectrum was randomly generated in between the two bounds, thus allowing to take into account of a certain scatter given by the real train operation. Each of the load spectra was generated so as to correspond to a mileage N_{LS} of nearly 30000 km on railway lines.

Based on the LEFM assumption for which $K \propto \sigma\sqrt{\pi a}$, once a K_a value is calculated for a given stress value σ_a, considering the same geometry, a K_b can be calculated by the simple linear relationship:

$$K_b = K_a(\sigma_b/\sigma_a) \tag{8}$$

Therefore, the K values related to each stress value were calculated according to Eq. (8) in such a way to obtain the K for all the different stress amplitudes without any further simulation.

A user made algorithm was built up to perform fatigue life predictions for the axle and the considered procedure can be schematically described with Fig. 10.

Each *n-th* load spectrum built up in between the bounds defined in Fig. 9 consisted of a sequence of stress amplitudes σ_j, each of them associated with a given number of cycles n_j, the latter defining the amount of repetitions that the axle undergoes at a given stress level σ_j. For each extraction of the stochastic procedure, a *n-th* load spectrum $\left(n_j, \sigma_j\right)$ was randomly generated. The main NASGRO equation (Eq. (2)) was rearranged in the form of Eq. (9) so as to calculate the corresponding Δa_j:

$$\Delta a = \sum_j \left\{ n_j * C \left[\left(\frac{1-f}{1-R} \right) \Delta K_j \right]^m \frac{\left(1 - \Delta K_{th}/\Delta K_j\right)^p}{\left\{1 - \Delta K_j/[(1-R)K_c]\right\}^q} \right\} \tag{9}$$

where Δa represents the crack advance produced after the n-th single load spectrum application. It is worth noting that, in Eq. (8), ΔK_j was calculated for the σ_j of the block by means of the above defined linear relationship (Eq. (8)). Such Δa was then used to calculate the amount of repetitions of the n-th load spectrum that need to be applied to achieve a crack advance from a_i to a_{i+1}. Namely, Eq. (9) allowed to calculate the number of spectra that can be travelled for each given crack advance $\Delta a_{i,i+1} = a_{i+1} - a_i$ by means of Eq. (10):

$$\Delta N_{i,i+1} = \frac{\Delta a_{i,i+1}}{\Delta a} N_{LS} \tag{10}$$

where N_{LS} is the total mileage of a single spectrum application. Very small discrete advances $\Delta a_{i,i+1}$ were defined within the procedure (1–2% per step) so as to guarantee that predictions were independent from the $\Delta a_{i,i+1}$. Finally, summing up all the $\Delta N_{i,i+1}$ from the initial crack depth of 2 mm and up to the failure when crack depth reaches half diameter, the total fatigue life ΔN was obtained. ΔN were then converted in corresponding mileages by considering a train wheel having an average railway wheel diameter D_{rw} equal to 940 mm.

5. Stochastic procedure description

Fatigue crack-growth of mechanical components is generally modelled by using material parameters and loading conditions derived in a deterministic way. However, the fracture phenomenon is inherently stochastic and the solutions obtained deterministically are limiting outcomes that generally need to be subsequently associated with relatively large safety margins. On the other hand, an approach based on a probabilistic standpoint is capable of providing outcomes not only in terms of a unique mean value or based on a best-fit curve of experimental data, but in terms of both mean value and dispersion around it. From a more general point of view, a stochastic approach should be as simple as possible while maintaining a reasonable accuracy for

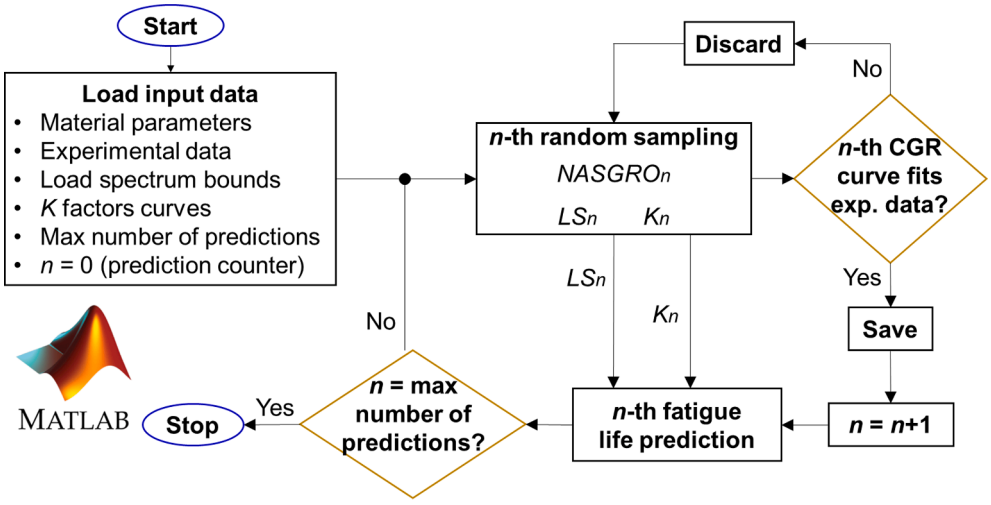


Fig. 11. Flow chart of the stochastic approach developed in the MATLAB routine.

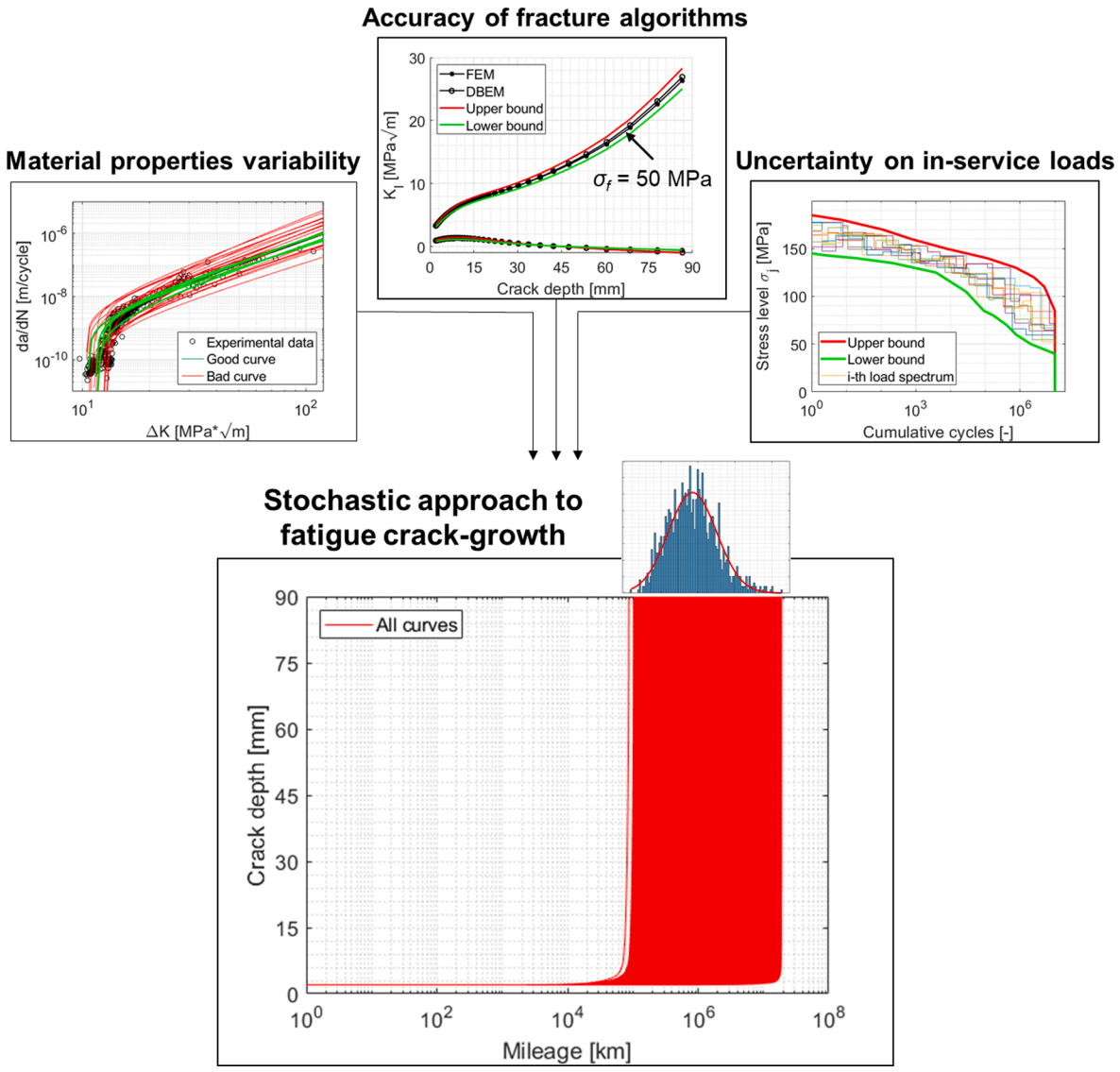


Fig. 12. Block diagram illustrating the proposed fatigue crack-growth stochastic procedure.

calculating the fatigue lives of components in presence of defects.

In this work, the procedure for the stochastic assessment of fatigue crack-growth phenomenon in a railway axle was fully performed within a user-made MATLAB [55] routine. The logic implemented within the routine is illustrated in the flowchart in Fig. 11. The routine considers as input data the random variables presented in the previous Sections 2–4 and provides as outputs the residual fatigue lives estimated for the axle. Inputs are considered as fully random and uncorrelated and no particular, e.g. normal/lognormal, probability density functions, were expressly enforced. With reference to Fig. 11, the procedure can be detailed in the following substeps:

1. the procedure starts by reading the required input data, i.e. material parameters and their ranges of variability (Table 3), experimental data (in Fig. 1), upper/lower bounds for the K curves (Fig. 8) and for the load spectra generation (Fig. 9), the maximum user defined number of predictions to perform;
2. for each n-th extraction, a set of NASGRO parameters $\left(C_n, m_n, p_n, \Delta K_{th,0,n}\right)$ were randomly sampled among their allowable values and used to generate a n-th CGR curve; this latter is compared with all the experimental data obtaining two RMS_n errors (Eq. (5)) so as to evaluate whether or not the n-th CGR curve well fits the experimental data;
3. only for the accepted sets of NASGRO parameters, one load spectrum curve LS_n and two K values curves K_n (one for bending, one for press-fit) are generated between their upper/lower bounds;
4. the so obtained LS_n, K_n and CGR curves were used to obtain a n-th fatigue life prediction;
5. the procedure is iterated until the user-defined maximum number of predictions to perform is reached.

Standing the considered sampling strategy, the maximum number of predictions to perform had to be sufficiently high in such a way to guarantee that the obtained results were not varying appreciably anymore. The maximum number of predictions to perform was set up as equal to 2e6 so as to guarantee that the variation of both mean value and standard deviation of the residual mileages were lower than 1e-8. The requested number of predictions turned out to be noteworthy, however relatively short runtimes were achieved (<24 h) avoiding the need to resort to more advanced sampling algorithms, e.g. Latin hypercube sampling as made in [21,56].

The abovementioned procedure was also illustrated in the block diagram of Fig. 12. As shown in Figs. 11 and 12, the random nature of the

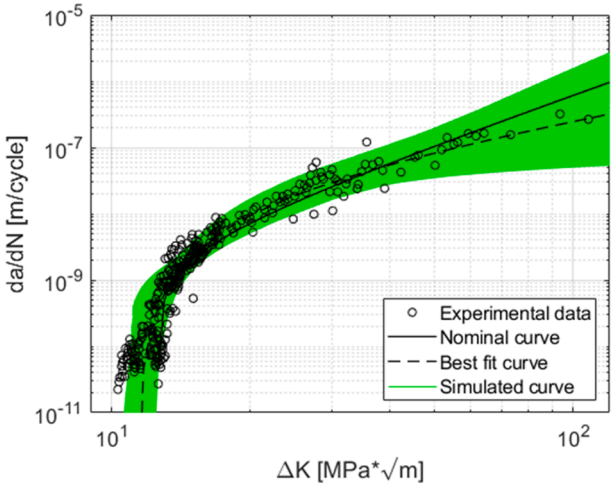


Fig. 13. CGR diagram comprising the experimental data, all the simulated curves, with best-fit and nominal curves highlighted.

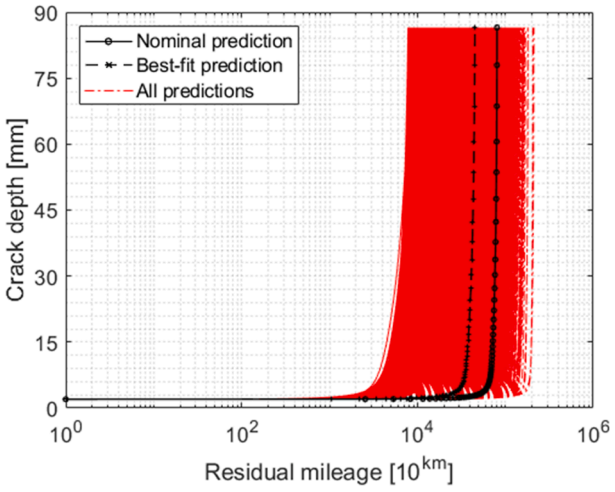


Fig. 14. Fatigue life predictions for all the simulations; predictions obtained by using best-fit and nominal curves are highlighted.

fatigue crack-growth in the railway axle was quantified in order to achieve a probabilistic description taking into account of the variabilities given by the material properties, the actual in-service loads and the crack-growth algorithm accuracy. With such an approach, the distribution of the fatigue life predictions with allowance for these sources of uncertainty were obtained, thus leading to an improved and more robust control over the safety required by these critical mechanical components.

6. Results and discussion

The presented procedure aimed to provide a robust tool capable to perform probabilistic fatigue life predictions for a railway axle (Fig. 12) by taking into account the most typical sources of uncertainty. Through the framework detailed in Section 5, fed with the input variables developed in Sections 2–4, it was possible to quantitatively evaluate the distribution of residual fatigue lives predicted for the axle. Even though calculations were performed with all the data sets, some figures reported below comprised a reduced number of data sets to improve their readability.

All the calculations were completed in nearly 24 h on a common laptop (4-th series i7 CPU, 16 GB RAM). All the valid CGR curves are shown in Fig. 13, superimposed to the experimental data, the nominal

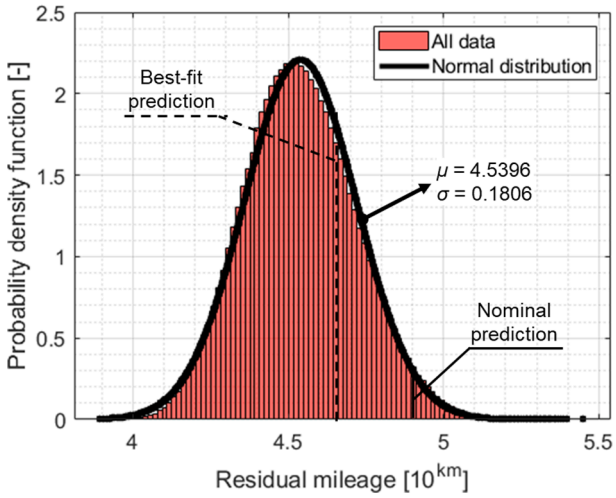


Fig. 15. Probability density function for the predicted residual mileages.

and the best-fit CGR curves; this latter is the CGR curve that provided the lowest RMS errors. It can be noticed a good correlation for both near threshold regime and Paris' range, while a worse correlation was observed in the critical region, where a lower amount of material data was available. This does not represent an issue because no appreciable impact on the life predictions is expected at such high ΔK values. All the estimated residual fatigue lives were reported in Fig. 14, whereas the final residual mileages were also reported in Fig. 15 as a distribution of occurrences.

Fig. 15 shows that the calculated predictions can be by means of a lognormal distribution in a very good way. Mileages presented a large lifespan distribution, ranging from 1e4 km to 2e5 km. Both best-fit and nominal CGR curves produced undoubtedly non-conservative predictions since respectively 74 or 97% of predictions turned out to be lower than them. This outcome clearly demonstrated how a single deterministic calculation, even if representing a best-fit of experimental data, can lead to non-conservative evaluations. The need to resort to a more robust stochastic approach for fatigue crack-growth assessments appears to be obvious.

The correlation matrix for the NASGRO parameters C, m, p, $\Delta K_{th,0}$ is reported in Fig. 16. The main diagonal shows the occurrences of the four parameters, that do not follow a normal/lognormal distribution. These impositions are typically and arbitrarily enforced a priori in several works [10,11,17,20,21,23] and they do not seem to be replicated by the here computed more realistic shapes. Out of the main diagonal, it can be observed that the material parameters are mostly uncorrelated, with the exception of C and m that provided an evident linear relationship. This linear dependence was well known from literature, as reported in dedicated works [57,58].

Fig. 17 reported the CGR curves obtained by averaging all the NASGRO parameters that provided the 3% either lowest or highest life predictions. These curves allowed to understand how either conservative or non-conservative predictions can be obtained when fitting CGR experimental data. Lowest predictions can be obtained by reducing the $\Delta K_{th,0}$ and by increasing the initial slope of the CGR curve, i.e. increasing C and reducing m. These modifications allow to increase the crack propagation velocity at low crack sizes, in turn producing a reduction in the fatigue lifetime. This outcome provides a reference useful to drive deterministic fatigue crack-growth methodologies in calculating conservative/non-conservative predictions.

Same concept was also applied to highlight the load spectrum effect. Fig. 18 reported all the simulated load spectra, in addition to two load spectra obtained by averaging all the spectra that provided the 3% highest and lowest predictions. A null impact of the high stress levels on the life predictions was observed, especially in terms of stresses but also

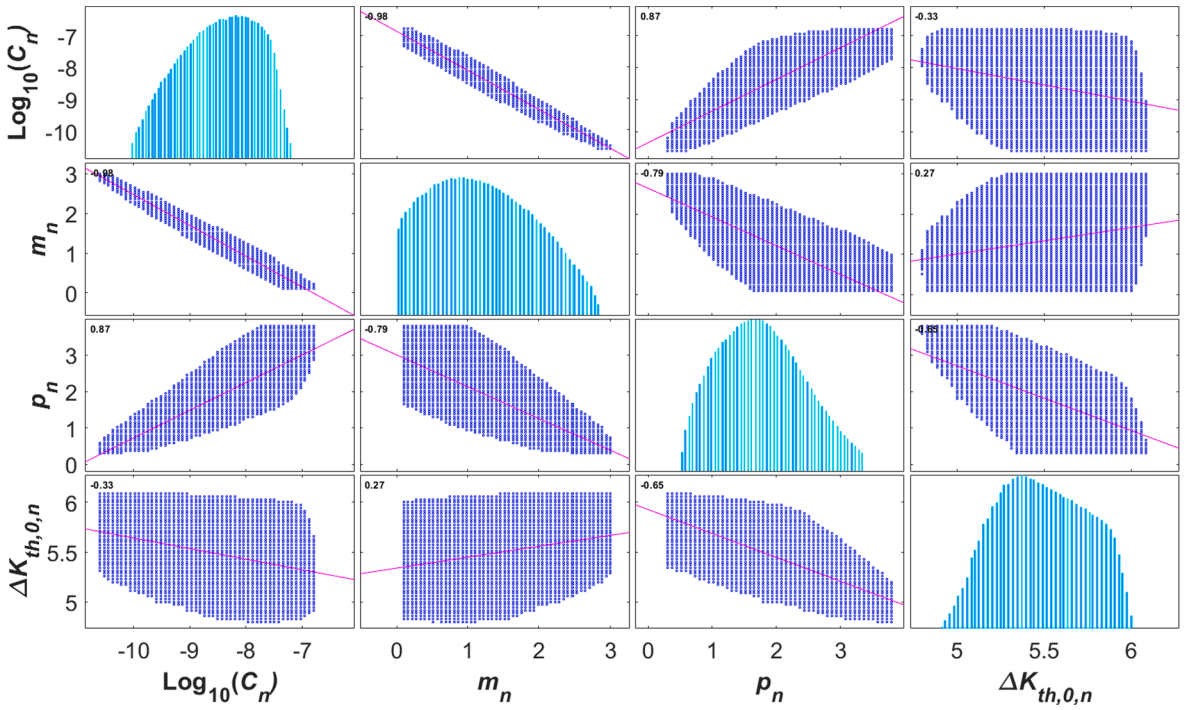


Fig. 16. Correlation matrix for NASGRO parameters C, m, p, $\Delta K_{th,0}$.

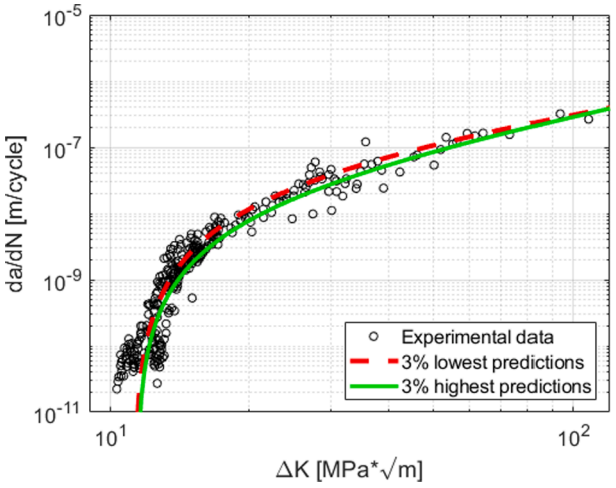


Fig. 17. CGR curve that provided conservative/non-conservative predictions.

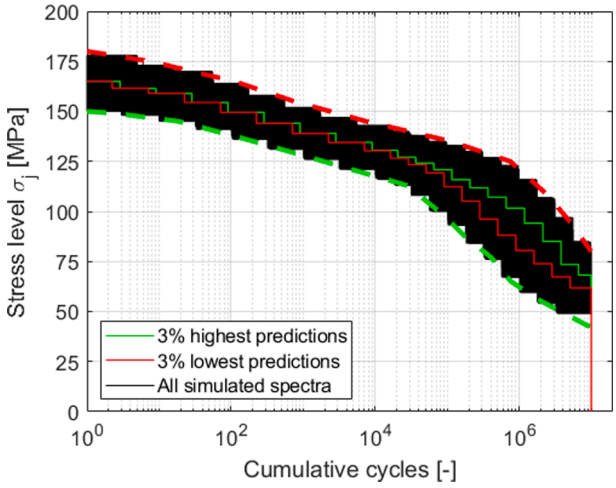


Fig. 18. All the simulated load spectra; highlight of average load spectra that provided conservative/non-conservative predictions.

in terms of cycles. Instead, considering the low stress levels, these stress amplitudes turned out to play a relevant impact with the largest contribute given by their cycles (i.e. by the "shape" of the load spectrum). It was also known from literature [59,60], that the different parts of the load spectrum can affect the fatigue life depending on the initial considered crack size. For the here considered 2 mm initial crack depth, the low stress levels produced a $\Delta K < \Delta K_{th}$ initially, therefore producing a null crack-growth, i.e. many cycles were run with a null cost for residual life. Different scenarios can be envisaged for larger initial crack sizes.

Finally, being obvious that the higher the K values are, the lower the life prediction is, the overall impact of this source of variability on the lifespan was simply quantified and reported in Fig. 19. It was observed that lifespan distribution was more affected by the accurate K evaluation rather than the load spectrum variability. This outcome is expected to be even more evident when considering more complex geometries, for which a higher discrepancy in K values can be expected for various methodologies.

7. Conclusions

This manuscript reported a stochastic approach to evaluate the phenomenon of fatigue crack-growth in a railway axle considering typical sources of uncertainty, such as: scattering of material data, actual in-service loading fluctuations, elements of approximation provided by the fracture algorithms.

The presented framework was based on numerical and experimental data that provided the input parameters for the presented approach, returning as outputs the probability distribution of the residual fatigue life of the axles in presence of defects. These results demonstrate the limitations of deterministic fatigue crack-growth calculations providing

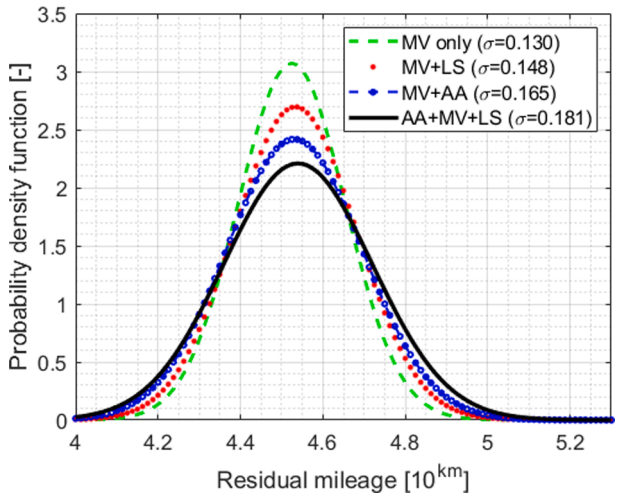


Fig. 19. Probability density functions for the predicted residual mileages under various input variabilities: "MV", "LS" and "AA" stand for Material Variability, Load Spectrum effects and Accuracy of fracture Algorithms respectively.

a reference tool to better drive these approaches to improve their assessments.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] Citarella R. Residual strength evaluation by DBEM for a cracked lap joint. Frattura ed Integrità Strutturale 2015;35:523–33. https://doi.org/10.3221/IGF-ESIS.35.58.

[2] Gallegos Mayorga L, Sire S, Correia JAFO, De Jesus AMP, Rebelo C, Fernández-Canteli A, et al. Statistical evaluation of fatigue strength of double shear riveted connections and crack growth rates of materials from old bridges. Eng Fract Mech 2017;185:241–57. https://doi.org/10.1016/j.engfracmech.2017.05.039.

[3] ISO EN 1993, Eurocode 3: design of steel structures; 2005.

[4] ASME BPVDC III-Div. 1, ASME boiler & pressure vessel design code – III Div. 1 – rules for construction of nuclear facility components; 2005.

[5] W.J. O'Donnell, Code design and evaluation for cyclic loading – sections III and VIII. In: Rao K, editor. Companion Guid to ASME Boil Press Vessel code, 3rd ed., vol. 2. New York: ASME; 2009. Doi: 10.1115/1.802191.ch39.

[6] Citarella R, Giannella V, Vivo E, Mazzeo M. FEM-DBEM approach for crack propagation in a low pressure aeroengine turbine vane segment. Theor Appl Fract Mech 2016;86:143–52. https://doi.org/10.1016/j.tafmec.2016.05.004.

[7] Giannella V, Fellinger J, Perrella M, Citarella R. Fatigue life assessment in lateral support element of a magnet for nuclear fusion experiment "Wendelstein 7-X". Eng Fract Mech 2017;178:243. https://doi.org/10.1016/j.engfracmech.2017.04.033.

[8] Grell WA, Laz PJ. Probabilistic fatigue life prediction using AFGROW and accounting for material variability. Int J Fatigue 2010;32(7):1042–9. https://doi.org/10.1016/j.ijfatigue.2009.12.001.

[9] Guida M, Penta F. A Bayesian analysis of fatigue data. Struct Saf 2010;32(1):64–76. https://doi.org/10.1016/j.strusafe.2009.08.001.

[10] Beretta S, Foletti S, Rusconi E, Riva A, Socie D. A log-normal format for failure probability under LCF: Concept, validation and definition of design curve. Int J Fatigue 2016;82:2–11. https://doi.org/10.1016/j.ijfatigue.2015.08.027.

[11] Citarella R, Apicella A. Advanced Design Concepts and Maintenance by Integrated Risk Evaluation for Aerostructures. Struct Durab Health Monit 2006;2(3):183–96. https://doi.org/10.3970/SDHM.2006.002.183.

[12] Klinger C, Bettge D. Axle fracture of an ICE3 high speed train. Eng Fail Anal 2013; 35:66–81. https://doi.org/10.1016/j.engfailanal.2012.11.008.

[13] Beretta S, Bruni S, Cinieri E, Corazza G, Curti G, De Iorio A, et al. Relazione incidente ferroviario di Viareggio 29 giugno 2009. In: Penal Proceedings 60305/09, Court of Lucca; October 2011.

[14] Lonsdale C, Stone D. North American axle failure experience. In: International Seminar on Railway Axles, Imperial College, London, 2003. DOI: 10.1243/0954409043125941.

[15] European Standard EN13103. Railway application – Wheelsets and bogies - Non-powered axles – Design method. Brussels, Belgium: European Committee for Standardization (CEN); 2001.

[16] European Standard EN13104. Railway application – Wheelsets and bogies - Powered axles – Design method. Brussels, Belgium: European Committee for Standardization (CEN); 2001.

[17] Beretta S, Carboni M. Application of fatigue crack growth algorithms to railway axles and comparison of two steel grades. Proc. Instn Mech. Engr. Part F: J. Rail and Rapid Transit 218 2004. https://doi.org/10.1243/0954409043125888.

[18] Zerbst U, Vormwald M, Andersch C, Mädler K, Pfuff M. The development of a damage tolerance concept for railway components and its demonstration for a railway axle. Eng Fract Mech 2005;72:209–39. https://doi.org/10.1016/j.engfracmech.2003.11.011.

[19] Beretta S, Regazzi D. Probabilistic fatigue assessment for railway axles and derivation of a simple format for damage calculations. Int J Fatigue 2016;86:13–23. https://doi.org/10.1016/j.ijfatigue.2015.08.010.

[20] Beretta S, Carboni M. Experiments and stochastic model for propagation lifetime of railway axles. Eng Fract Mech 2006;73:2627–41. https://doi.org/10.1016/j.engfracmech.2006.04.024.

[21] Zhu SP, Foletti S, Beretta S. Probabilistic framework for multiaxial LCF assessment under material variability. Int J Fatigue 2017;103:371–85. https://doi.org/10.1016/j.ijfatigue.2017.06.019.

[22] Beretta S, Carboni M, Fiore G, Lo Conte A. Corrosion–fatigue of A1N railway axle steel exposed to rainwater. Int J Fatigue 2010;32:952–61. https://doi.org/10.1016/j.ijfatigue.2009.08.003.

[23] Larsen JM, Jha SK, Szczepanski CJ, Caton MJ, John R, Rosenberger AH, et al. Reducing uncertainty in fatigue life limits of turbine engine alloys. Int J Fatigue 2013;57:103–12. https://doi.org/10.1016/j.ijfatigue.2013.01.012.

[24] UNI 6787:1971, "Assi fucinati di acciaio speciale legato, bonificati, ad elevate caratteristiche di fatica e di tenacità per sale montate di rotabili ferroviari. Qualità, prescrizioni e prove.", Italian Standardization Body UNI; 1971.

[25] Newman JC. A Crack opening stress equation for fatigue crack growth. Int J Fract 1984;24:R131–5. https://doi.org/10.1007/BF00020751.

[26] Behavior of Surface and Corner Cracks Subjected to Tensile and Bending Loads in Ti-6Al-4V Alloy. In: Ernst HA, Saxena A, McDowell DL, editors. Fracture Mechanics: Twentysecond Symposium, vol. 1, ASTM STP 1131. American Society for Testing and Materials, Philadelphia; 1992. p. 519–46.

[27] NASGRO® Fracture Mechanics & Fatigue Crack Growth Analysis Software, 2019; website: http://www.nasgro.swri.org.

[28] Elber W. Fatigue crack closure under cyclic tension. Eng Fract Mech 1970;2:37–45. https://doi.org/10.1016/0013-7944(70)90028-7.

[29] Elber W. The significance of fatigue crack closure. In: Rosenfeld M, editor. Damage Tolerance in Aircraft Structures. West Conshohocken, PA: ASTM International; 1971. p. 230–42. Doi: 10.1520/STP26680S.

[30] El-Haddad MH, Smith KN, Topper TH. Fatigue crack propagation of short crack. Trans. ASME, J Engng Mater Technol 1979;101:42–6. https://doi.org/10.1115/1.3443647.

[31] James LA, Mills WJ. Review and synthesis of stress intensity factor solutions applicable to cracks in bolts. Eng Fract Mech 1988;30(5):641–54. https://doi.org/10.1016/0013-7944(88)90156-7.

[32] Murakami Y. Stress intensity factors. United States: N. p., 1987. Web. https://www.osti.gov/biblio/5798454.

[33] Giannella V, Citarella R, Perrella M, Shlyannikov V. Surface crack modelling in an engine compressor disc. Theor Appl Fract Mech 2019;103:102279. https://doi.org/10.1016/j.tafmec.2019.102279.

[34] Giannella V, Dhondt G, Kontermann C, Citarella R. Combined static-cyclic multiaxial crack propagation in cruciform specimens. Int J Fatigue June 2019;123:296–307. https://doi.org/10.1016/j.tafmec.2019.102279.

[35] Citarella R, Giannella V, Lepore M, Dhondt G. Dual boundary element method and finite element method for mixed-mode crack propagation simulations in a cracked hollow shaft. Fatigue Fract Eng Mater Struct 2018;41(1):84–98. https://doi.org/10.1111/ffe.12655.

[36] Giannella V, Perrella M, Citarella R. Efficient FEM-DBEM coupled approach for crack propagation simulations. Theor Appl Fract Mech 2017;91:76–85. https://doi.org/10.1016/j.tafmec.2017.04.003.

[37] Citarella R, Cricrì G, Armentani E. Multiple crack propagation with Dual Boundary Element Method in stiffened and reinforced full scale aeronautic panels. Key Eng Mater 2013;560:129–55. https://doi.org/10.4028/www.scientific.net/KEM.560.129.

[38] Citarella R. Non Linear MSD crack growth by DBEM for a riveted aeronautic reinforcement. Adv Eng Softw 2009;40(4):253–9. https://doi.org/10.1016/j.advengsoft.2008.04.007.

[39] Citarella R. MSD Crack propagation on a repaired aeronautic panel by DBEM. Adv Eng Softw 2011;42(10):887–901. https://doi.org/10.1016/j.advengsoft.2011.02.014.

[40] Zhuang X, Augarde C, Bordas S. Accurate fracture modelling using meshless methods, the visibility criterion and level sets: Formulation and 2D modelling. Numer Methods Eng 2011;86(2):249–68. https://doi.org/10.1002/nme.3063.

[41] Bobaru F, Foster JT, Geubelle PH, Silling SA. Handbook of Peridynamic Modeling. CRC Press/Taylor and Francis Group; 2017. ISBN: 9781482230437.

[42] FRANC3D, Reference manual v6. Fracture Analysis Consultants Inc.; 2011.

[43] BEASY V10r18, Documentation. C.M. BEASY Ltd., 2018.

[44] WIDEM EU Project, Wheel-set Integrated Design and Effective Maintenance; 2007. Web: http://www.widem.org.

[45] European standard. Railway applications, wheel sets and bogies, axles, product requirement. EN 13261; 2003. p. 1–51.

[46] Dhondt G. General behaviour of collapsed 8-node 2-D and 20-node 3-D isoparametric elements. Int J Numer Methods Eng 36:1223–43. Doi: 10.1002/nme.1620360708.

[47] Giannella V, Perrella M. Multi-axial fatigue numerical crack propagation in cruciform specimens. Frattura ed Integrità Strutturale 2019;13(48):639–47. https://doi.org/10.3221/IGF-ESIS.48.61.

[48] Rigby RH, Aliabadi MH. Mixed-mode J-integral method for analysis of 3D fracture problems using DBEM. Eng Anal Bound Elem 1993;11:239–56. https://doi.org/10.1016/0955-7997(93)90026-H.
[49] Rigby RH, Aliabadi MH. Decomposition of the mixed-mode J-integral revisited. Int J Solids Struct 1998;35(17):2073–99. https://doi.org/10.1016/S0020-7683(97)00171-6.
[50] Wawrzynek PA, Carter B, Banks-Sills L. The M-integral for computing stress intensity factors in generally anisotropic materials. NASA/CR-2005-214006; 2005.
[51] Sih GC. Strain energy density factor applied to mixed mode crack problems. Int J Fract 1974;10:305–21. https://doi.org/10.1007/BF00035493.
[52] Sih GC, Cha BCK. A fracture criterion for three-dimensional crack problems. Eng Fract Mech 1974;6(4):699–723. https://doi.org/10.1016/0013-7944(74)90068-X.
[53] Erdogan F, Sih GC. On the extension of plates under plane loading and transverse shear. J Basic Eng 1963;85D(4):519–25. https://doi.org/10.1115/1.3656897.
[54] Lütkepohl K, Esderts A, Luke M, Varfolomeev I. Sicherer und wirtschaftlicher. Betrieb von Eisenbahnfahrwerken 2009. https://doi.org/10.2314/GBV:614268583.
[55] MATLAB, R2020a. Natick, Massachusetts: The MathWorks Inc.; 2019.
[56] Giannella V, Lombardi R, Pisani MM, Federico L, Barbarino M, Citarella R. A novel optimization framework to replicate the vibro-acoustics response of an aircraft fuselage. Appl Sci 2020;10:2473. https://doi.org/10.3390/app10072473.
[57] Cortie MB, Garrett GG. On the correlation between the C and m in the Paris equation for fatigue crack propagation. Eng Frac Mech 1988;30(1):49–58. https://doi.org/10.1016/0013-7944(88)90254-8.
[58] Bergner F, Zouhar G. A new approach to the correlation between the coefficient and the exponent in the power law equation of fatigue crack growth. Int J Fatigue 2000;22(3):229–39. https://doi.org/10.1016/S0142-1123(99)00123-1.
[59] Pokorný P, Náhlík L, Hutař P. Comparison of different load spectra on residual fatigue lifetime of railway axle. Procedia Eng 2014;74:313–6. https://doi.org/10.1016/j.proeng.2014.06.269.
[60] Pokorný P, Hutař P, Náhlík L. Residual fatigue lifetime estimation of railway axles for various loading spectra. Theor Appl Fract Mech 2016;82:25–32. https://doi.org/10.1016/j.tafmec.2015.06.007.